A380X ATC COM and datalink

The MFD ATC COM pages and the SD mailbox of the A380 FCOM on the FBW datalink, with a datalink that connects reliably, ATC route clearances and constraints loaded into SEC 3, the OIT COMPANY COM inbox and the MFD System Message Area

Aircraft	A380X (the ACARS provider connection is shared with the A32NX)
Where to find it	MFD system menu **ATC COM** (or the KCCU ATC COM key); the SD **MAILBOX**; MFD FMS **INIT** (FLT NBR); flyPad > Settings > ATSU / AOC; OIT COMPANY COM; MFD header
Branch	`core/a380x-atccom-datalink`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). `core/printer` builds on it.
Status	ATC COM pages and mailbox: tested in the sim (ATIS, frames, fixes), logon and MAYDAY, PAN PAN PAN. Flight number connection and SEND FAILED: OK. ACARS provider: seen working with BeyondATC. Mailbox FILE FULL fix: OK; @ markers: OK. LOAD-SEC3: route clearance OK; crossing constraints not sim-tested. COMPANY COM inbox and MFD System Message Area: not sim-tested, unit tests only.
Pictures	Captain MFD and SD in the local browser harness, running the build installed in the sim; the messages come from the harness (no ATC network). flyPad page in the same harness. No pictures for the mailbox fixes and LOAD-SEC3.

1. ATC COM pages and the SD mailbox
2. Datalink connection with the flight number, and SEND FAILED
3. ACARS provider connection
4. SD mailbox fixes: FILE FULL after CANCEL, uplink @ markers
5. LOAD-SEC3: ATC route clearances and crossing constraints into SEC 3
6. OIT COMPANY COM inbox with the real AOC messages
7. MFD System Message Area: CHECK FMS MESSAGE, NEW ATIS RECEIVED, ATIS SYS MSG

1. ATC COM pages and the SD mailbox

> **Aircraft:** A380X | **Where:** MFD ATC COM pages; SD MAILBOX | **Status:** Tested in the sim by the user: ATIS, REQUEST / OTHER REPORTS / EMERGENCY frames and the fixes of that test (frame delete, label layout, font sizes, ATIS time): OK. Logon to BeyondATC (RJJJ), MAYDAY sent, PAN PAN PAN sent. Answering uplinks of a live controller (WILCO, UNABLE...) not recorded. Unit tests: 33.

What it is for

The A380 talks to ATC by datalink: the crew logs on to a center, sends requests and reports, receives clearances and answers them, and gets the airport ATIS as text. The FCOM describes the MFD **ATC COM** pages (DSC-46-10-20-30) and the SD **MAILBOX** (DSC-46-10-10-70). They are all built on their FCOM figures and connected to the FBW ATSU, the same datalink system as the A32NX DCDU.

> **NOTE**
>
> The network is chosen in the flyPad: Settings > ATSU / AOC (Hoppie, BeyondATC or SayIntentions). The datalink needs VHF3 in DATA mode on the RMP.

Page (header)	Use
CONNECT	Logon (NOTIFICATION), CONNECTION STATUS with ADS, MAX UPLINK DELAY.
REQUEST	Build a request from frames (climb, direct, offset, weather deviation, speed, clearances...) and send it to the mailbox.
REPORT & MODIFY	Position report, other reports, modify a prepared answer.

MSG RECORD	The record of the closed dialogues.
ATIS	Request and read the ATIS of three airports.
EMER	MAYDAY / PANPAN and the emergency reports.

Logging on: CONNECT

1. CONNECT > NOTIFICATION: type the **ATC CENTER** (4 letters) and press **NOTIFY**. The page shows NOTIFYING, then the center in NOTIFIED TO CENTERS (with the time), or NOTIFICATION FAILED.
2. When ATC accepts, CONNECTION STATUS shows the ACTIVE ATC (and the NEXT ATC after a transfer). DISCONNECT ends the connection.

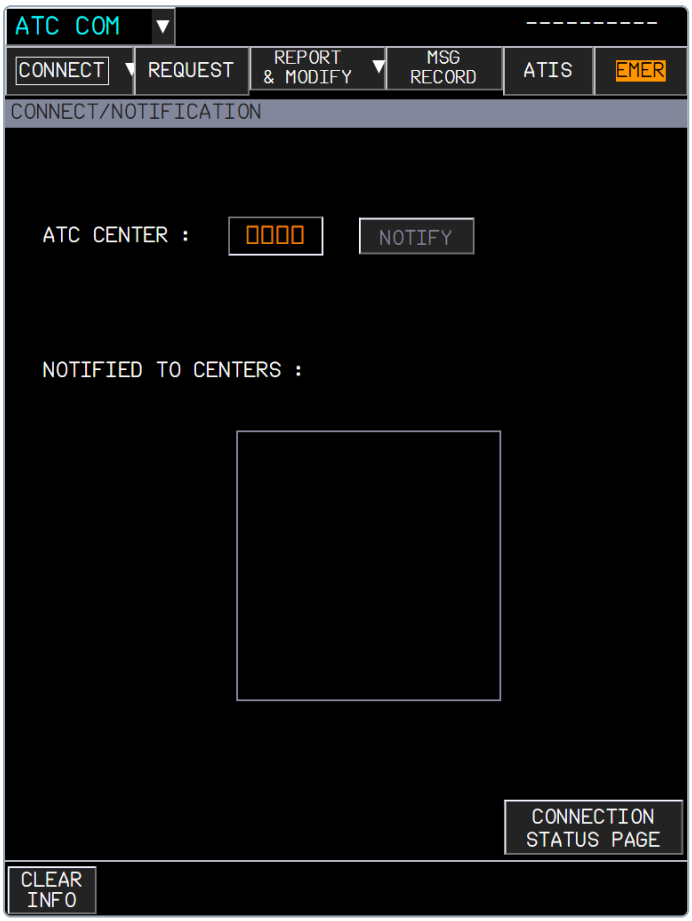


Figure 1. CONNECT / NOTIFICATION.

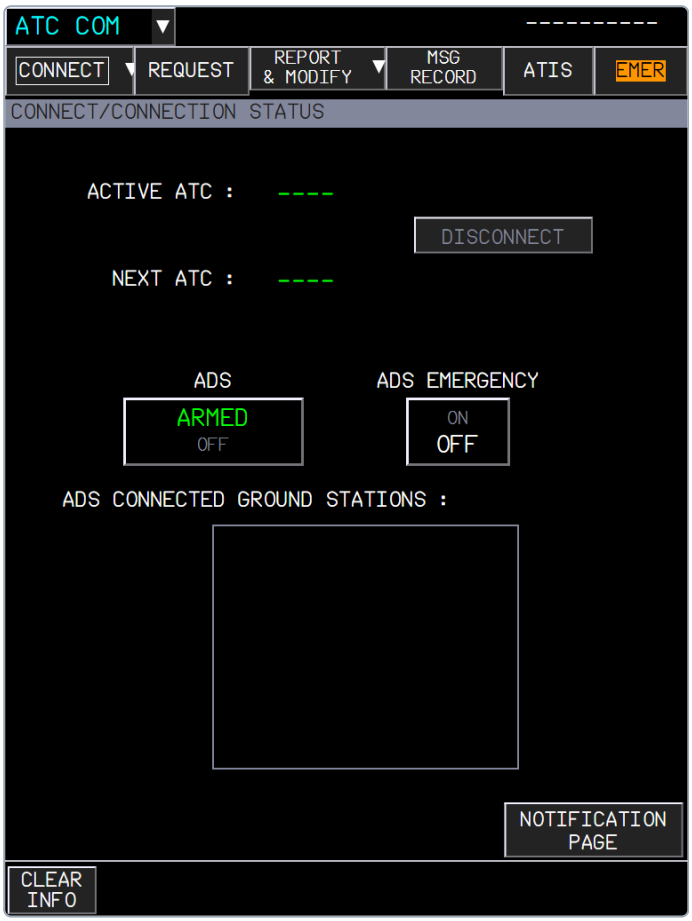


Figure 2. CONNECTION STATUS: active and next ATC, ADS ARMED / OFF, ADS EMERGENCY.

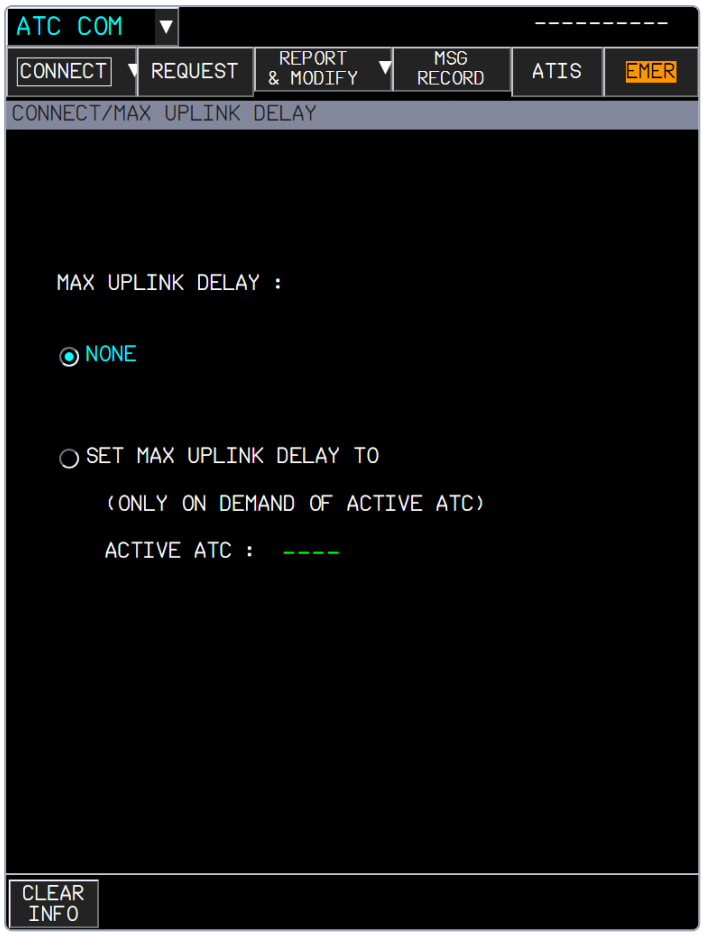


Figure 3. MAX UPLINK DELAY: NONE, or a delay set on the demand of the active ATC.

Asking ATC: REQUEST

1. Open the menus on the right (VERTICAL, LATERAL, SPEED, CLEARANCE, WHEN CAN WE EXPECT, OTHER) and pick the frames you need; up to 5 frames, each with its fields.
2. Fill the fields (FCOM entry formats; amber boxes are mandatory). ADD TEXT adds free text. The bin deletes a frame.
3. Press **XFR TO MAILBOX**: the request goes to the SD mailbox, where you **SEND** it.

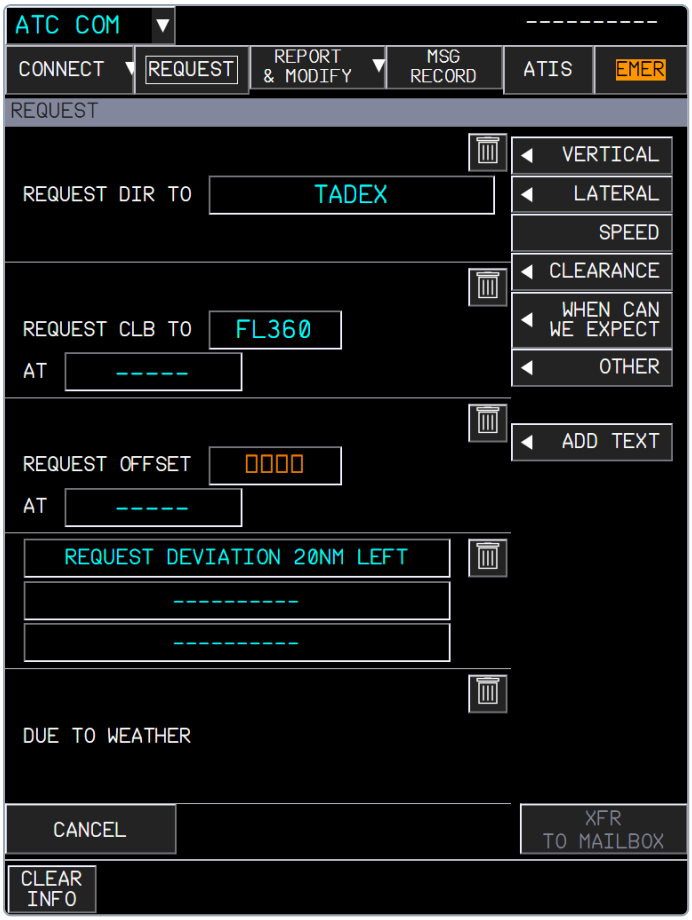


Figure 4. A request of 5 frames: DIR TO TADEX, CLB TO FL360, OFFSET (amber: to fill), free text, DUE TO WEATHER.

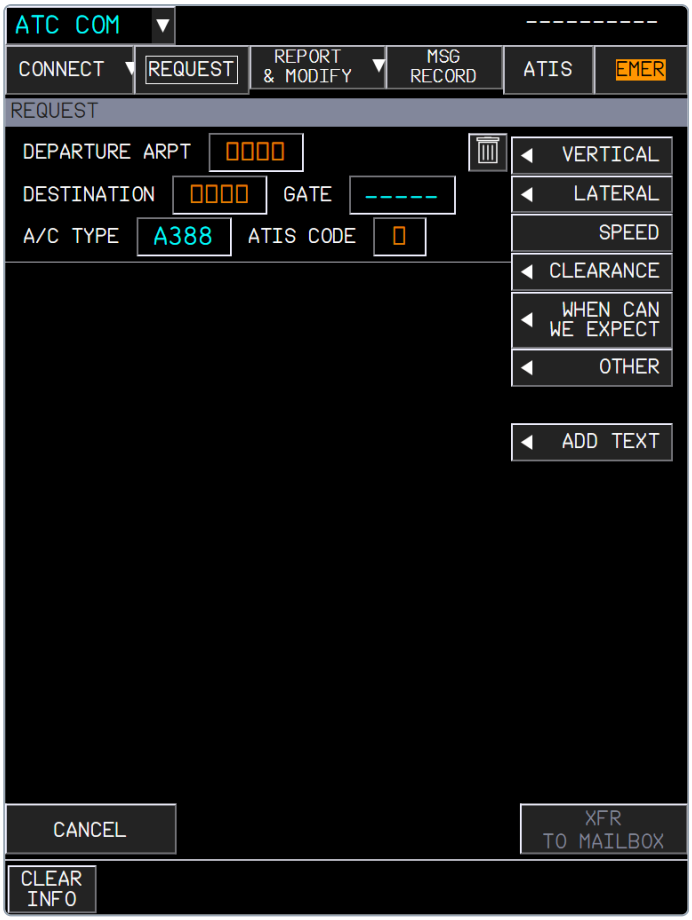


Figure 5. CLEARANCE > DEPARTURE: the departure clearance request (DCL): departure airport, destination, gate, aircraft type, ATIS code.

Reports: REPORT & MODIFY

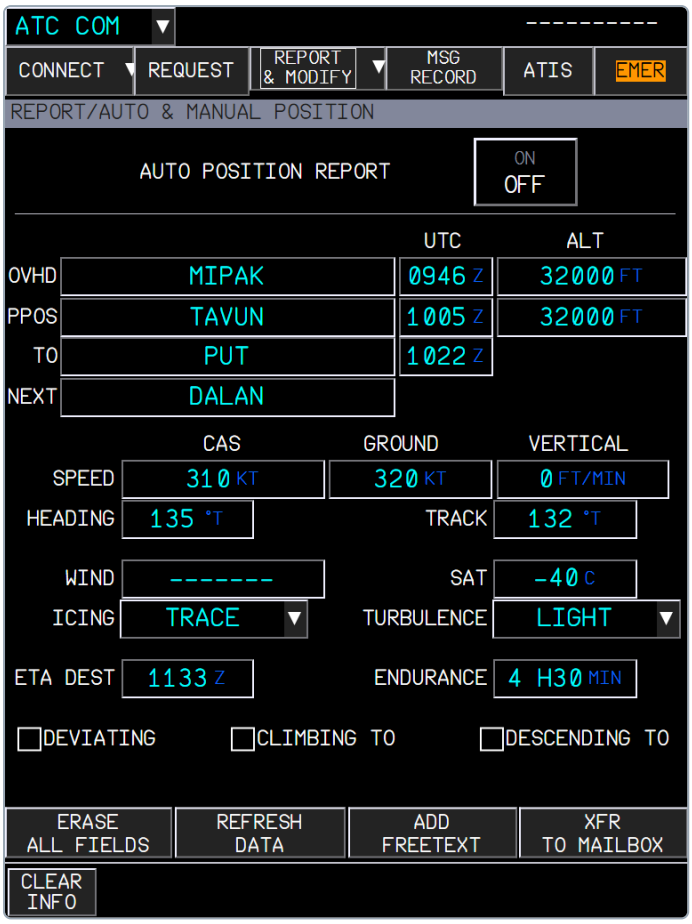


Figure 6. AUTO & MANUAL POSITION: the position report filled by the FMS (overhead, present position, to, next, times, levels, speeds, wind, SAT, ETA, endurance); AUTO POSITION REPORT ON sends it at each reporting point.

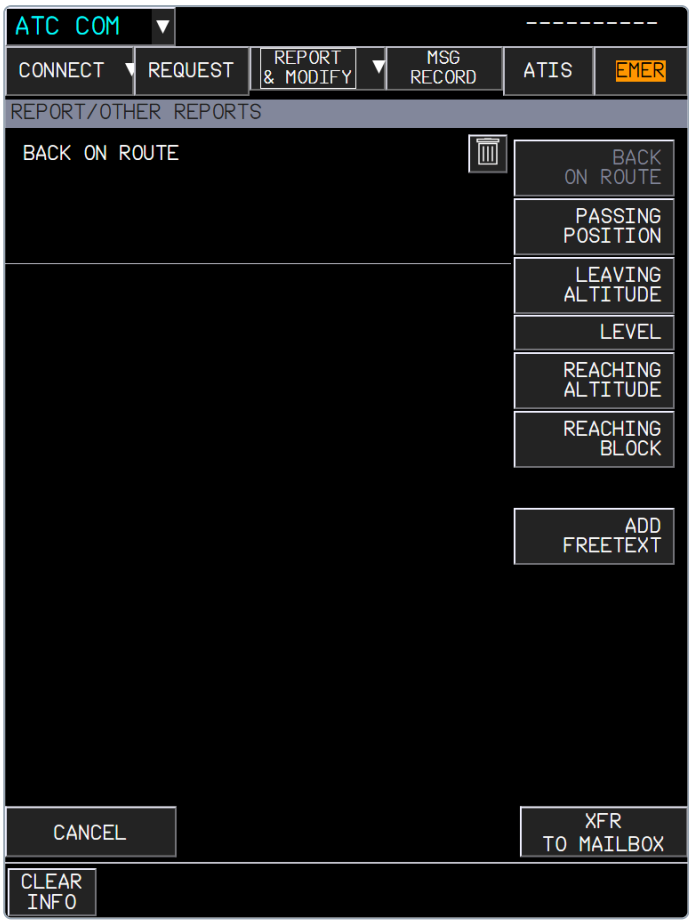


Figure 7. OTHER REPORTS: back on route, passing position, leaving / reaching altitude, level, reaching block, free text.

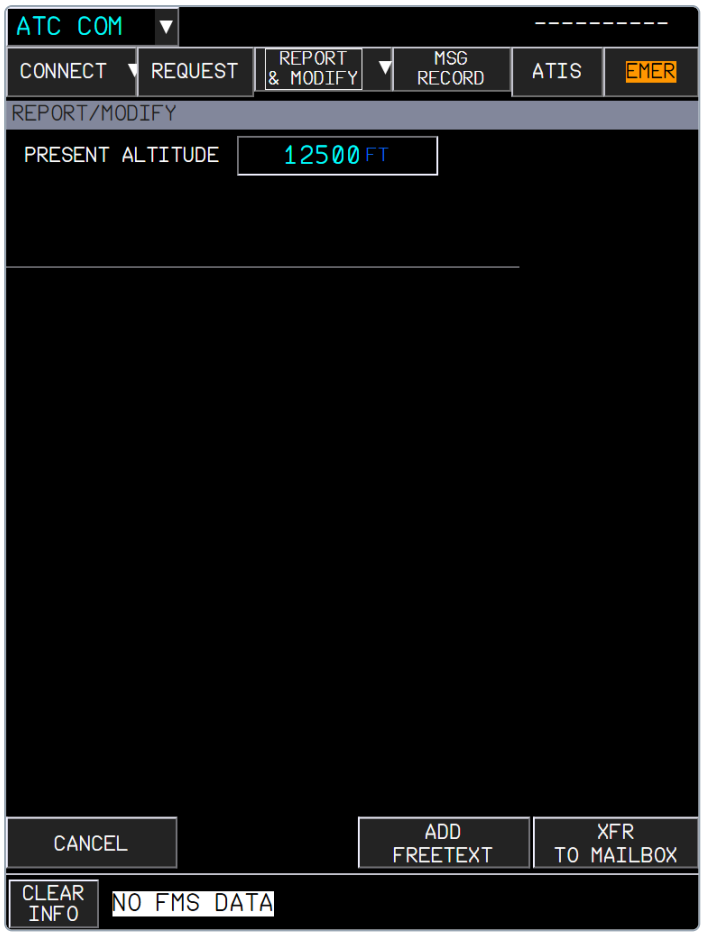


Figure 8. MODIFY: a prepared answer (here PRESENT ALTITUDE 12500 FT) that the crew can change before sending.

The SD MAILBOX

Every message to and from ATC goes through the mailbox at the bottom of the SD. It shows the active ATC, the ADS status, the message and the answer buttons the message expects.

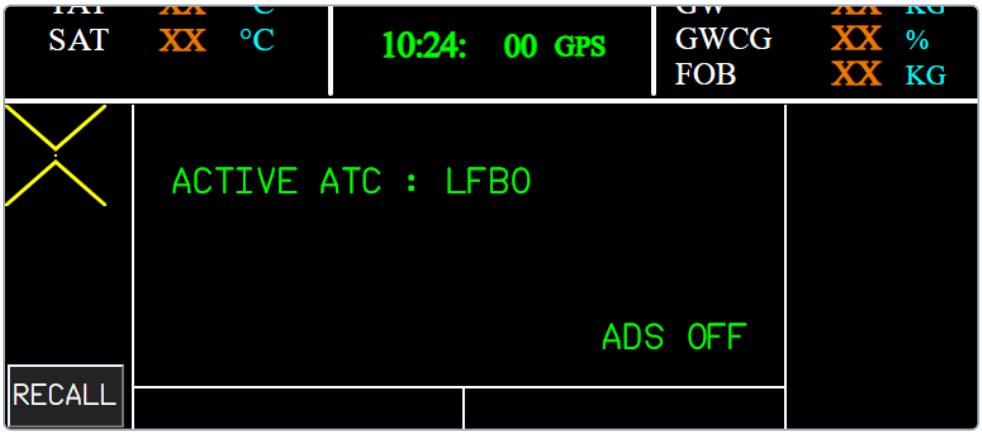


Figure 9. Mailbox without a message: ACTIVE ATC LFBO, ADS OFF.

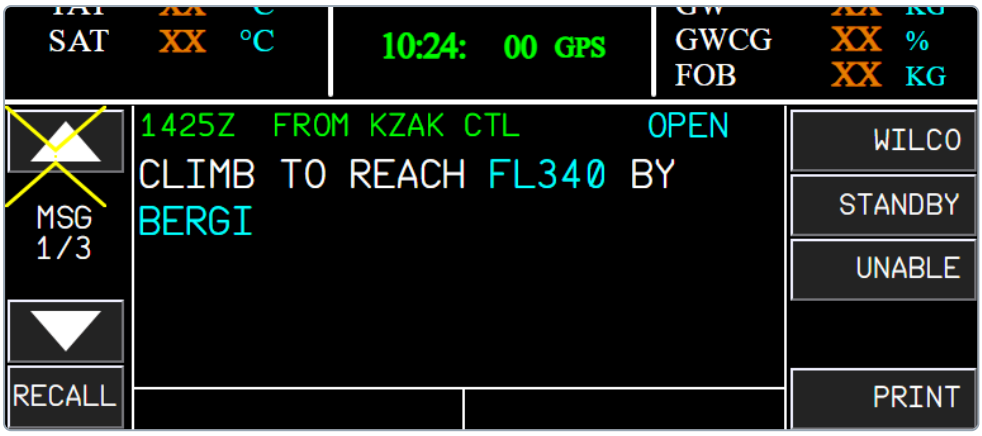


Figure 10. An uplink: CLIMB TO REACH FL340 BY BERGI, OPEN, with WILCO / STANDBY / UNABLE. MSG 1/3: three messages are waiting.

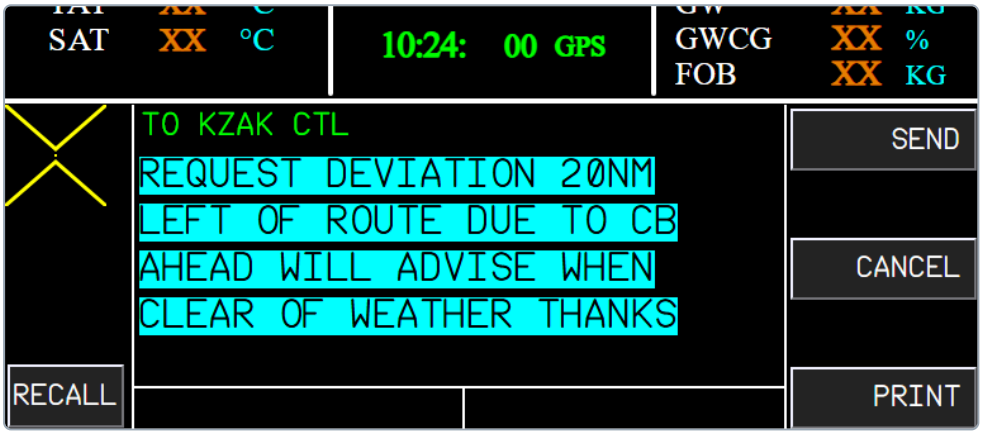


Figure 11. A request ready to go: SEND or CANCEL.

The message expects	Buttons
WILCO / UNABLE (a clearance)	WILCO, STANDBY, UNABLE
AFFIRM / NEGATIVE (a question)	AFFIRM, STANDBY, NEGATIVE
ROGER (information)	ROGER
A departure or oceanic clearance	ACK / REFUSE

UNABLE and NEGATIVE can add a reason (DUE TO WEATHER, A/C PERFORMANCE, TURBULENCE) and free text; MODIFY opens the answer on the MFD. The message colours follow the FCOM status legend.

MSG RECORD and ATIS

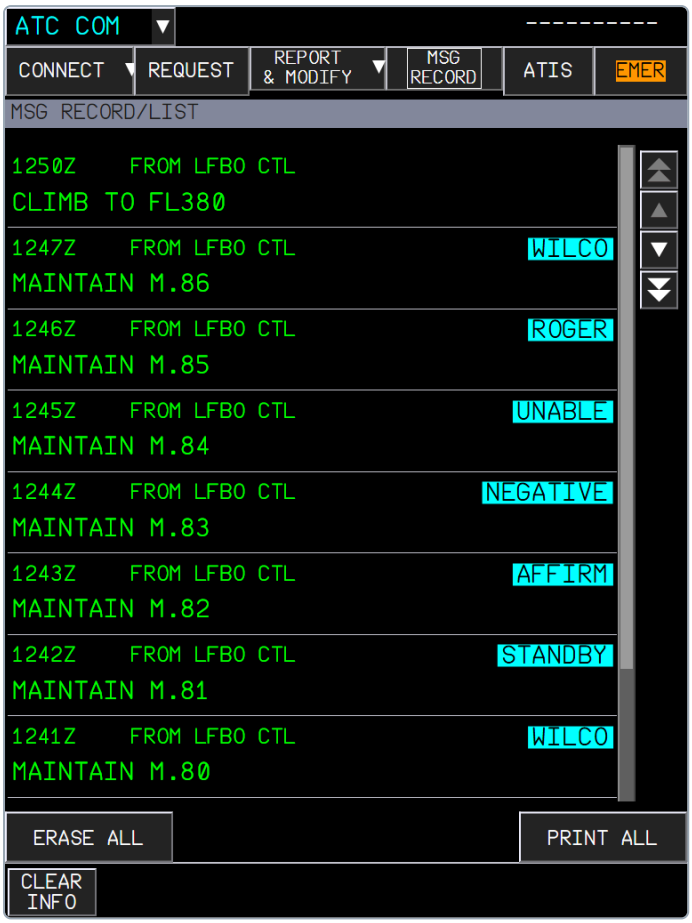


Figure 12. MSG RECORD: the closed dialogues with their time and the answer given; ZOOM shows one in full; ERASE ALL, PRINT ALL.

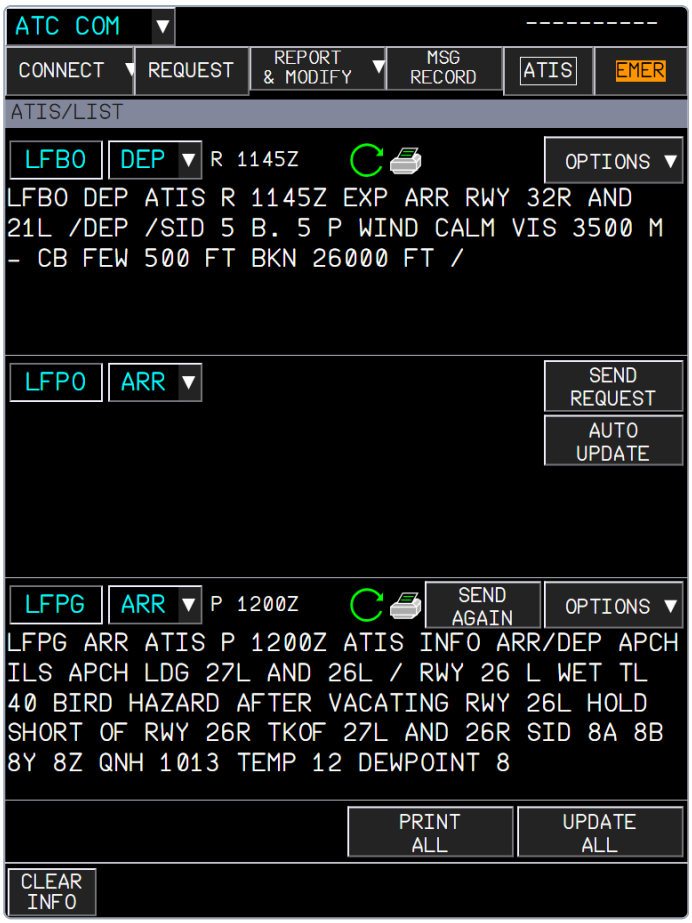


Figure 13. ATIS: three request areas (departure, arrival, alternate, following the FMS airports), each with its ATIS text, AUTO UPDATE, and SEND REQUEST / SEND AGAIN.



Figure 14. ATIS RECEIVED: the full text of an ATIS, RETURN TO LIST.

Emergency: EMER

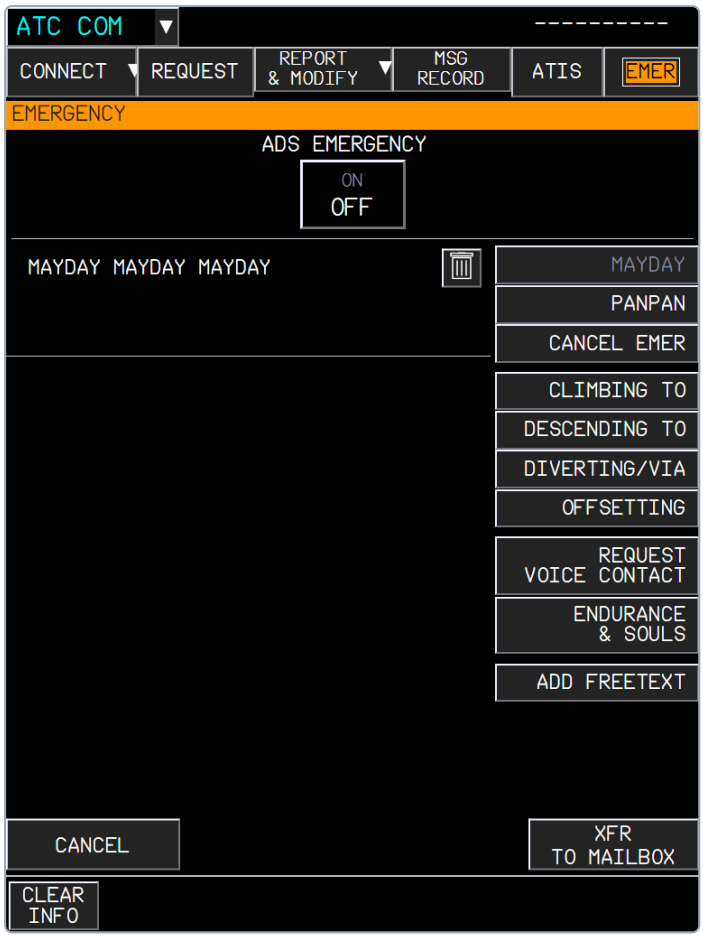


Figure 15. EMERGENCY (orange title): MAYDAY MAYDAY MAYDAY frame, PANPAN, CANCEL EMER and the emergency reports (climbing / descending to, diverting, offsetting, voice contact, endurance and souls). MAYDAY also switches the ADS to EMERGENCY.

What to expect

Situation	What you should see
NOTIFY with no network set	NOTIFICATION FAILED.
Request with an empty mandatory field	XFR TO MAILBOX stays greyed.
Uplink received	Mailbox message with its answer buttons.
Answer sent	The dialogue closes and goes to MSG RECORD.
ATIS requested	The text appears in its area; a newer version replaces it (AUTO UPDATE).
MAYDAY sent	ADS EMERGENCY ON.

Limitations and why

Limitation	Why	What to do
Needs an online datalink network.	ATC messages come from Hoppie, BeyondATC or SayIntentions through the flyPad ATSU settings.	Set the network in the flyPad.
No real ADS contracts.	The ATSU networks do not handle ADS-C; ADS ARMED / EMERGENCY is shown locally.	-
On this branch alone, every ATC COM PRINT says PRINTER NOT AVAIL.	Printing (ATIS, MSG RECORD, mailbox) is on `core/printer`, which builds on this branch.	-

Answering the uplinks of a live controller is not recorded as tested.	Tested in the sim: logon, MAYDAY and PAN PAN downlinks, the ATIS, a route clearance uplink (LOAD-SEC3). The answer buttons were checked with harness messages.	Report any wrong answer element.

> **NOTE**
> The A380X README (list of what is not implemented, ATC / datalink line) now says that the MFD ATC COM pages and the SD mailbox work through the FBW ATSU and its ACARS providers, and that the datalink functions outside those pages are not implemented.

How to use

1. Set the network in the flyPad (Settings > ATSU / AOC), VHF3 in DATA mode on the RMP, and your network callsign as FLT NBR on the MFD FMS INIT page.
2. MFD system menu > **ATC COM** > CONNECT > NOTIFICATION: type the ATC CENTER and press **NOTIFY**.
3. To ask ATC: REQUEST, pick the frames, fill them, **XFR TO MAILBOX**, then **SEND** in the SD mailbox.
4. To answer ATC: use the buttons of the message in the SD mailbox (WILCO / UNABLE, AFFIRM / NEGATIVE, ROGER...).
5. For the ATIS: ATIS page, airport of the request area, SEND REQUEST; tick AUTO UPDATE to get the new versions.
6. In an emergency: EMER, MAYDAY or PANPAN frame, XFR TO MAILBOX, SEND.

How to test

#	Do this	Expected result
1	A380X, any phase. flyPad > Settings > ATSU / AOC: ACARS provider set (Hoppie with its logon code, or BeyondATC running); RMP VHF3 in DATA; MFD FMS INIT: FLT NBR = your network callsign. Open MFD system menu > **ATC COM**.	Header CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER; the REQUEST page opens.
2	CONNECT > NOTIFICATION: type a center (with BeyondATC: the one in its Frequencies tab, COM tuned to it), press **NOTIFY**.	NOTIFYING, then the center in NOTIFIED TO CENTERS with the time (or NOTIFICATION FAILED). Once ATC accepts, CONNECTION STATUS shows the ACTIVE ATC and the SD mailbox shows it too.
3	ATIS page: in the first request area type an airport, press **SEND REQUEST**.	The ATIS text in the area; its time is the time written in the ATIS (e.g. 1151Z), ---- when it has none. Opening it shows ATIS RECEIVED with the full text.
4	REQUEST: VERTICAL menu > a climb frame, type FL360; add a second frame and delete it with the bin symbol.	The labels and fields of a line follow each other without overlap; the bin removes the frame.
5	Press **XFR TO MAILBOX**.	The request in the SD mailbox with SEND and CANCEL.
6	Press **SEND** in the mailbox (logged on).	The request is sent; when the dialogue closes it goes to MSG RECORD with its time.
7	EMER page: MAYDAY MAYDAY MAYDAY frame, XFR TO MAILBOX, SEND.	The MAYDAY is sent and CONNECT shows ADS EMERGENCY. BeyondATC answers NO MESSAGE HANDLING AVAILABLE FOR THIS REQUEST (not supported on its side).
8	Receive an uplink (from a live controller, or self-sent, see LOAD-SEC3 below) and answer it.	The message in the mailbox with the buttons it expects; after the answer the dialogue closes and goes to MSG RECORD.

2. Datalink connection with the flight number, and SEND FAILED

> **Aircraft:** A380X | **Where:** MFD FMS INIT (FLT NBR); FMS message area; SD MAILBOX | **Status:** Tested in the sim by the user: logon to BeyondATC after the FLT NBR fix OK; flight number connection from the FMS and SEND FAILED without ATC OK. FMS DATALINK NOT AVAIL not recorded as seen in the sim. Unit tests: 5.

What it is for

The INIT page FLT NBR field wrote the flight plan only: the field did not refresh and the entry never connected the datalink networks (FBW TELEX and the ACARS provider), so ATC logons and downlinks failed. And a message sent from the mailbox with no ATC center was dropped without a word. Now the FMS connects the networks whenever the flight number of the active flight plan changes, and the mailbox says SEND FAILED (A380 FCOM DSC-46-10-20-60).

What you see

The ATC COM pages send and receive through the aircraft datalink, which connects with the flight number of the active flight plan (FMS INIT, FLT NBR). It must be your callsign on the network (Hoppie, BeyondATC or SayIntentions: the one set in the flyPad, Settings > ATSU / AOC).

- The flight number connects the datalink however it gets there: typed on the INIT page, from the company flight plan (SimBrief), from an activated secondary flight plan, or already there when you reload the flight.
- If the connection is refused, or the datalink does not answer, the FMS message area shows **FMS DATALINK NOT AVAIL** (white). It goes away by itself once connected; typing the flight number again tries again.
- A message sent from the mailbox with no ATC center connected (no logon done, or the center logged you off) shows **SEND FAILED** instead of nothing.

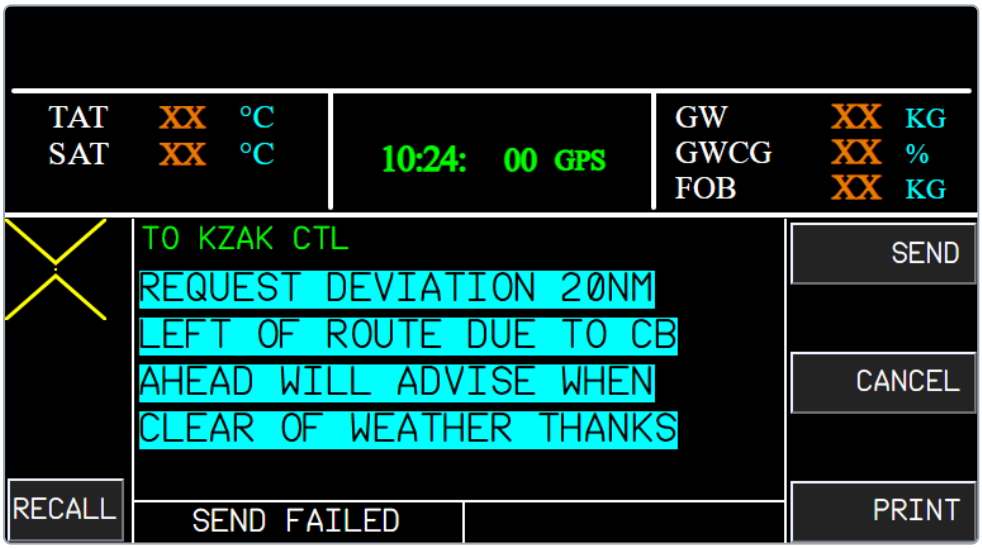


Figure 16. SEND FAILED under a message the mailbox could not send (no active ATC center).

> **NOTE**
> With BeyondATC: log on to the center shown in its Frequencies tab (e.g. RJJJ), above FL145. BeyondATC does not handle MAYDAY or PAN PAN by datalink: it answers NO MESSAGE HANDLING AVAILABLE FOR THIS REQUEST. Declare an emergency by voice with it.

How to use

1. Enter the flight number on the MFD FMS **INIT** page (FLT NBR), or get it with the company flight plan. It must be your callsign on the network.
2. If **FMS DATALINK NOT AVAIL** shows, check the ACARS provider, then type the flight number again to retry.
3. Log on (ATC COM > CONNECT) before sending: without an active ATC the mailbox says SEND FAILED.

How to test

#	Do this	Expected result
1	A380X on the ground, ACARS provider running and set in flyPad > Settings > ATSU / AOC. MFD FMS **INIT**: type the flight number in FLT NBR.	The field shows the new flight number at once. flyPad ATSU / AOC: the provider shows Connected (green).
2	ATC COM > CONNECT > NOTIFICATION: NOTIFY a center.	The logon goes out (NOTIFYING, then NOTIFIED TO CENTERS): the networks are connected.
3	Load a flight with the flight number already in the active flight plan (or activate a SEC plan with a flight number).	The datalink connects without typing the number again.
4	Without logon (no ACTIVE ATC), prepare a request (REQUEST, XFR TO MAILBOX) and press **SEND** in the SD mailbox.	**SEND FAILED** under the message (before: nothing happened).
5	If the connection is refused (for example FBW TELEX refuses the flight number), look at the FMS message area.	**FMS DATALINK NOT AVAIL** in white; it goes away once connected, and typing the flight number again retries.

3. ACARS provider connection

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > ATSU / AOC | **Status:** Late start and close / restart of BeyondATC seen working in the user's flights; not re-checked after the final build. Unit tests: 4.

What it is for

The ATC and company datalink of the aircraft (CPDLC logon, requests, messages, ATIS and METAR) goes through an ACARS provider you choose on the flyPad: Hoppie, BeyondATC or SayIntentions. The aircraft must find it, then connect your flight number to it.

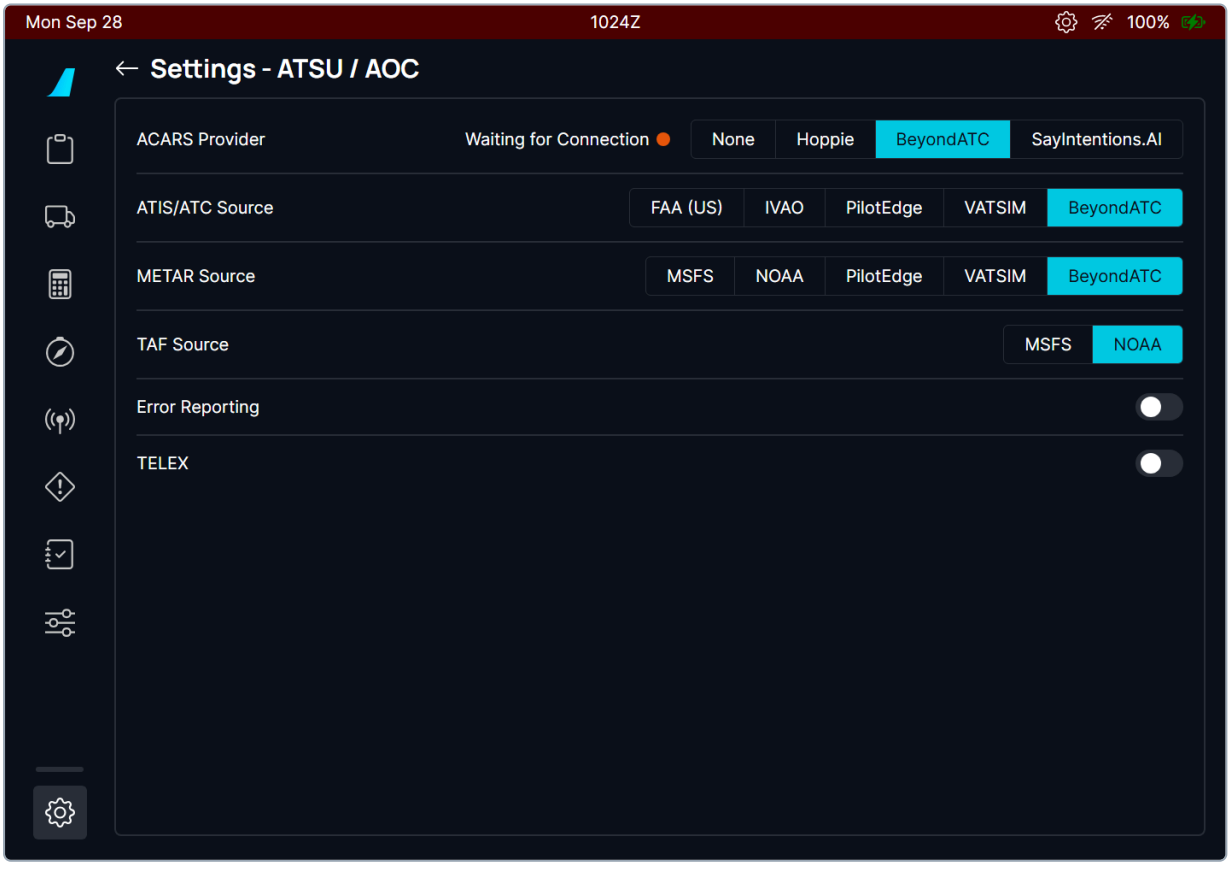


Figure 17. ACARS Provider: BeyondATC, "Waiting for Connection" until BeyondATC answers (then "Connected", green).

What happens now

- **Start the provider when you like.** The aircraft looks for it every 5 s for 5 minutes, then once a minute, without end. Your provider, ATIS and METAR settings are never changed.
- **Flight number.** Entered before the provider answers, it is connected as soon as it answers.
- **Provider closed.** Within about a minute the status goes back to "Waiting for Connection"; start it again and the aircraft connects your flight number again by itself. With BeyondATC, log on to the ATC center again (it forgets the logon when it closes).
- **TELEX on.** If FBW's TELEX service refuses your flight number, the ACARS provider is connected anyway (on the A380X, the FMS shows FMS DATALINK NOT AVAIL for the TELEX failure).

What to expect

- No extra traffic with Hoppie: once your flight number is connected, the regular polls (every 45 to 75 s) check the provider.
- Before this change, a provider started more than 5 minutes after the aircraft set the flyPad to NONE: select your provider again once if it shows NONE.

How to use

1. flyPad > Settings > **ATSU / AOC**: choose the ACARS provider (Hoppie, BeyondATC or SayIntentions).
2. Start the provider program whenever you like, before or after loading the aircraft.
3. Enter the flight number (MFD INIT on the A380X, MCDU INIT A on the A32NX): it is connected as soon as the provider answers.

How to test

#	Do this	Expected result
1	Load the aircraft with BeyondATC NOT running. flyPad > Settings > ATSU / AOC: ACARS provider BeyondATC.	"Waiting for Connection".
2	Wait more than 5 minutes, then start BeyondATC.	Within about a minute the status turns to "Connected" (green). The provider, ATIS and METAR settings are unchanged (not reset to NONE).
3	Enter the flight number before BeyondATC answers (restart the test with the flight number entered first).	The flight number is connected as soon as BeyondATC answers.
4	Close BeyondATC.	Within about a minute the status goes back to "Waiting for Connection".
5	Start BeyondATC again.	The aircraft connects the flight number again by itself; log on to the ATC center again (BeyondATC forgets the logon when it closes).

4. SD mailbox fixes: FILE FULL after CANCEL, uplink @ markers

> **Aircraft:** A380X | **Where:** SD MAILBOX; MFD ATC COM MSG RECORD | **Status:** Tested in the sim by the user: FILE FULL no longer sticks after cancels, PAN PAN PAN sent to BeyondATC; @ markers shown as cyan values.

What it is for

Two mailbox problems found in the sim with BeyondATC:

- **FILE FULL for good.** The mailbox keeps at most five open messages per direction and holds the next ones back with FILE FULL. A cancelled downlink (CANCEL of a prepared message) left the ATC function but kept its place in

the mailbox file, so after five cancels the file stayed full: every new downlink, a PAN PAN PAN included, waited behind FILE FULL. A cancel now frees the place, the held messages move in, and FILE FULL goes when the file has room again (A380 FCOM DSC-46: "File no more full").

- **Triangles around values.** Hoppie and BeyondATC mark the variable fields of a free text uplink with @ (CLRD TO @CYVR@ RWY @06R@), and the A380 display font draws @ as a triangle. The mailbox now shows the marked fields as main parameters, in blue (FCOM DSC-46-10-10-70), without the markers, as the A32NX DCDU does; an unclosed marker runs to the end of the text, and a word glued to a field ("@4611@.") stays glued. MSG RECORD drops the markers too.

What you should see

Situation	What you should see
Five downlinks prepared and cancelled, then a new one	The new downlink is shown with SEND / CANCEL; no FILE FULL.
More than five open downlinks	FILE FULL for the extra ones, until a place is freed (sent, closed or cancelled).
BeyondATC departure clearance "CLRD TO @CYVR@ RWY @06R@ ..."	CLRD TO CYVR RWY 06R ..., the values in blue, no triangles.

How to use

1. Nothing to set: CANCEL a prepared downlink as usual, read the uplinks as usual.

How to test

#	Do this	Expected result
1	A380X, logged on (ATC COM > CONNECT). Prepare a request (REQUEST, XFR TO MAILBOX) and press **CANCEL** in the SD mailbox. Do it six times.	Each new request shows SEND / CANCEL; FILE FULL never stays.
2	Then prepare an emergency (EMER, PANPAN frame, XFR TO MAILBOX) and SEND.	The PAN PAN PAN is sent (no FILE FULL).
3	With BeyondATC, ask for a departure clearance (REQUEST > CLEARANCE > DEPARTURE, XFR TO MAILBOX, SEND).	The clearance uplink in the mailbox: values (destination, runway, squawk...) in blue, no triangle symbols.
4	Close the dialogue and open ATC COM > **MSG RECORD**.	The recorded text has no @ characters.

5. LOAD-SEC3: ATC route clearances and crossing constraints into SEC 3

> **Aircraft:** A380X | **Where:** SD MAILBOX **LOAD-SEC3**; MFD FMS SEC INDEX | **Status:** Route clearance tested in the sim by the user (self-sent uplink): LOAD OK, ATC F-PLN INSERTED IN SEC 3. Crossing constraints and the REJECTED ATC INFO page not sim-tested. Unit tests: route clearance, constraints, mailbox logic.

What it is for

A route clearance or a crossing constraint received by CPDLC can be loaded into the FMS instead of being typed: the FCOM puts it in SEC 3, as a copy of the active flight plan with the clearance in it, for the crew to check and activate (A380 FCOM DSC-22-FMS-10-40, DSC-46-10-40). The mailbox has the **LOAD-SEC3** key in the 4th button position, on the uplinks that can be loaded only; it stays after WILCO, with CLOSE.

Uplink (CPDLC element)	What is loaded into SEC 3

Route clearance: CLEARED TO ... VIA ... (UM79), CLEARED ... (UM80), AT ... CLEARED ... (UM83)	SID (not once airborne), airways, waypoints, lat/long points, STAR and a new destination. The rest of the en route part is replaced; a UM79 route ends at its clearance limit.
Crossing constraints (UM46 to 53, 55, 56, 58 to 61): CROSS ... AT / AT OR ABOVE / AT OR BELOW / BETWEEN ..., AT ... speed, AT ... time	An altitude constraint, a speed constraint (90 kt to VMO 340 kt, in knots) or an RTA on that waypoint of the flight plan. The RTA is kept per secondary flight plan and follows it on a swap or an activation; the VERT REV RTA panel now works on a SEC plan.

What you should see

Situation	What you should see
LOAD-SEC3 pressed	Mailbox information area: LOADING, then LOAD OK, LOAD PARTIAL or LOAD FAILED.
Everything loaded	FMS message ATC F-PLN INSERTED IN SEC 3. SEC 3 holds the active flight plan with the clearance.
Some elements rejected	ATC F-PLN INSERTED IN SEC 3 / REJECTED INFO SEE SEC / INDEX. SEC INDEX, SEC 3: **REJECTED ATC INFO** button, listing each element (description, value, AT fix) and why: NOT IN DATABASE, AWY/WPT MISMATCH, DATA NOT ALLOWED IN PHASE, NOT IN F-PLN, FORMAT ERROR, ENTRY OUT OF RANGE.
Nothing usable	RECEIVED ATC MSG NOT VALID.
No FMS answer within 30 s, or a second LOAD-SEC3 while loading	LOAD NOT AVAIL.
A speed as a Mach number (e.g. M.82)	Rejected: FORMAT ERROR (the FCOM speed format is in knots).

> **NOTE**
> The constraint upload awaits in a plain statement, not as an operand: the sim's JavaScript engine can reject an await used as an operand and then blanks the whole MFD. No visible change.

Limitations and why

Limitation	Why	What to do
A duplicate waypoint ident is taken as the one nearest to the previous point.	Design choice: no NON UNIQUE rejection.	Check the route in SEC 3 before activating it.
A constraint on an en route waypoint with no climb or descent leg type is set as a descent constraint in cruise, else from the second half of the en route part.	The FMS needs CLB or DES for a constraint; this is a guess.	Check the constraint in SEC 3.
Needs an uplink of one of these CPDLC elements.	Live controllers rarely send them; BeyondATC route clearances were not seen.	Send yourself the uplink (How to test).

How to use

1. Receive the clearance in the SD mailbox and press **LOAD-SEC3**.
2. Wait for LOAD OK (or LOAD PARTIAL), then read the FMS message.
3. MFD FMS: SEC INDEX, SEC 3: check the flight plan (and REJECTED ATC INFO if shown).
4. Answer the uplink (WILCO / UNABLE) and, if you accept the clearance, activate SEC 3 (or swap it).

How to test

#	Do this	Expected result
1	A380X with an active flight plan whose en route part has waypoints ahead, datalink connected (flight number entered). Send yourself a route clearance. With BeyondATC (tested way, ask the owner of the setup first): open its local relay `http://localhost:57698/acars/system/connect.html` with `from=CZUL&to=<callsign>&type=cpdlc&packet=/data2/1//WU/CLEARED TO @<DEST>@ VIA @<FIX1> DCT <FIX2>@` (URL-encoded). With Hoppie, send the same CPDLC packet to your callsign through its connect.html.	Within a minute (next poll) the uplink is in the SD mailbox, the values in blue, with **LOAD-SEC3** in the 4th button position.
2	Press **LOAD-SEC3**.	LOADING, then LOAD OK in the mailbox; FMS message ATC F-PLN INSERTED IN SEC 3.
3	MFD FMS > SEC INDEX > SEC 3 F-PLN.	The active route with the clearance: the cleared fixes joined, the waypoints after them removed, the route ending at the clearance limit.
4	Send a clearance with a fix that does not exist (e.g. VIA @ZZZZZ@).	LOAD PARTIAL (or LOAD FAILED); SEC INDEX shows REJECTED ATC INFO with ZZZZZ and NOT IN DATABASE.
5	Send a crossing constraint on a waypoint ahead (CPDLC UM46, e.g. CROSS <FIX> AT FL240) and press LOAD-SEC3.	SEC 3: an altitude constraint FL240 on that waypoint (not sim-tested yet).
6	Press LOAD-SEC3 on a second uplink while the first one is still LOADING.	LOAD NOT AVAIL on the second one.

6. OIT COMPANY COM inbox with the real AOC messages

> **Aircraft:** A380X | **Where:** OIT (side console) AVNCS > COMPANY COM inbox; E/WD COMPANY MSG memo | **Status:** Not sim-tested (added; unit tests: CompanyComMailbox.spec).

What it is for

The OIT COMPANY COM inbox showed two hard-coded test messages. It now lists the company (AOC) messages received by the datalink, newest first, with an unread mark, the sender and the UTC time (A380 FCOM DSC-46-40-20, COMPANY COM application). Displaying a message marks it read; the COMPANY MSG memo goes once every message is read (FCOM: COMPANY MSG, "The flight crew received a company message")..

Situation	What you should see
A company message (for example a Hoppie free text) arrives	A new line at the top of the inbox with the unread mark; the E/WD shows COMPANY MSG.
The crew opens the message	The text shows; the line loses its unread mark.
Every message read	The COMPANY MSG memo goes.

Limitations and why

Limitation	Why	What to do
The other OIS FLT OPS applications are not done.	They are separate work.	-

How to use

1. Connect the datalink (flight number on MFD FMS INIT, AOC provider in flyPad > Settings > ATSU / AOC).
2. When COMPANY MSG shows, open the OIT COMPANY COM inbox and click the message to read it.

How to test

#	Do this	Expected result
1	A380X, datalink connected with Hoppie. Send yourself a free text telex from the Hoppie website.	The OIT COMPANY COM inbox lists it at the top, unread, with the sender and the UTC time; COMPANY MSG on the E/WD.
2	Click the message in the inbox.	The message shows; it is marked read and COMPANY MSG goes (if it was the only unread one).

What PR testers should check

- A Hoppie free text reaching the inbox, and COMPANY MSG cleared after reading it.
- Not sim-tested yet.

7. MFD System Message Area: CHECK FMS MESSAGE, NEW ATIS RECEIVED, ATIS SYS MSG

> **Aircraft:** A380X | **Where:** MFD header, all applications | **Status:** Not sim-tested (added; unit tests: MfdSystemMessageArea.spec, failing first).

What it is for

The A380 FCOM MFD header has a System Message Area that tells the crew about a message of another application while it is not shown (A380 FCOM MFD description, System Message Area). It now shows:

- **CHECK FMS MESSAGE**: an amber FMS message is pending while another application (ATC COM, SURV, FCU BKUP) is shown; removed when the crew opens an FMS page.
- **NEW ATIS RECEIVED**: a new ATIS report of the ATC COM datalink arrived while another application is shown; removed when the crew opens an ATC COM page.
- **ATIS SYS MSG**: an ATIS system message of the ATC centre (USE VOICE, SEND AGAIN, NO AUTO UPDATE, END OF UPDATE) while another application is shown; removed when the crew opens an ATC COM page.

Design choices: the area shows one message, the most recent one (the FCOM gives no priority); a message comes only from a new event, so going back to another application does not bring back a message already seen in its application.

Limitations and why

Limitation	Why	What to do
The KCCU MAILBOX key and the ATC MSG flashing are on `core/a380x-mailbox-du-reconf`.	They belong with the SD mailbox.	-

How to use

1. When a message shows in the MFD header, open the application it names (FMS, or ATC COM for the ATIS) to read it.

How to test

#	Do this	Expected result
1	A380X, MFD on ATC COM (or SURV). Cause an amber FMS message (any amber line of the FMS message list).	The MFD header shows **CHECK FMS MESSAGE**.
2	Datalink connected, MFD on the FMS. Request an ATIS on ATC COM, then go back to the FMS before it arrives.	When the ATIS arrives the header shows **NEW ATIS RECEIVED**.
3	Open the named application.	The header message goes.

What PR testers should check

- The layout of the message area in the header (fit, colour, position).
- Not sim-tested yet.